



Spectro Scientific

Confidence in knowing

Q5800 Expeditionary Fluid Analysis System

Customer Presentation

INNOVATIVE TECHNOLOGY TO EMPOWER
MAINTAINERS AT POINT OF USE

October 2014

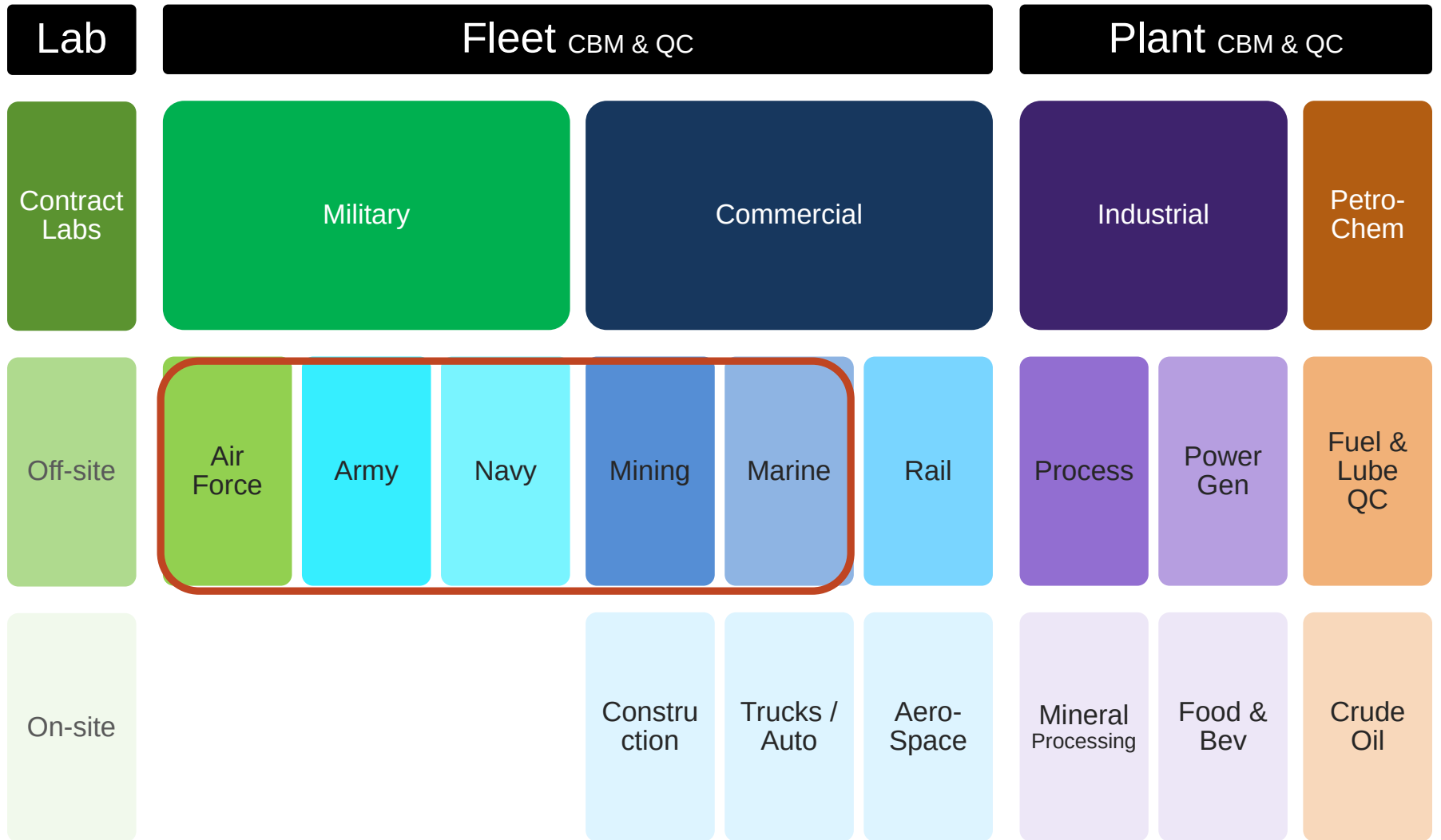


Agenda

- Expeditionary Fluid Analysis
- Current Challenges
- Q5800
- Technical Discussion
- Military Experience to date
- Other Projects
- Ordering and Configuration



Our Market Landscape

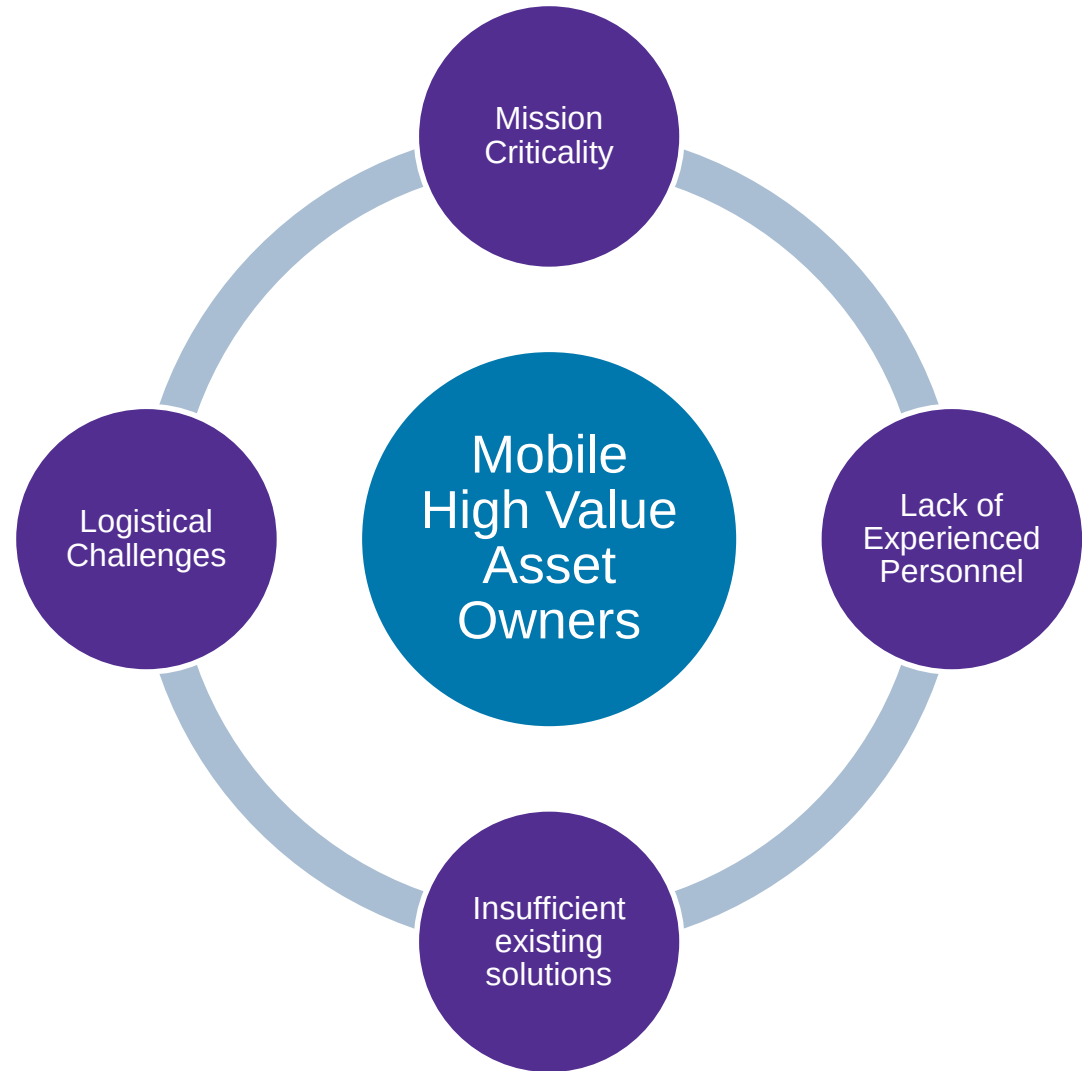


Past Success

Future Growth

An underserved community

- Mobile, Deployable Equipment
- Temporary, remote operating area
- Lack of experienced lab personnel
- Existing solutions not responsive – or catch



Why Expeditionary Fluid Analysis?

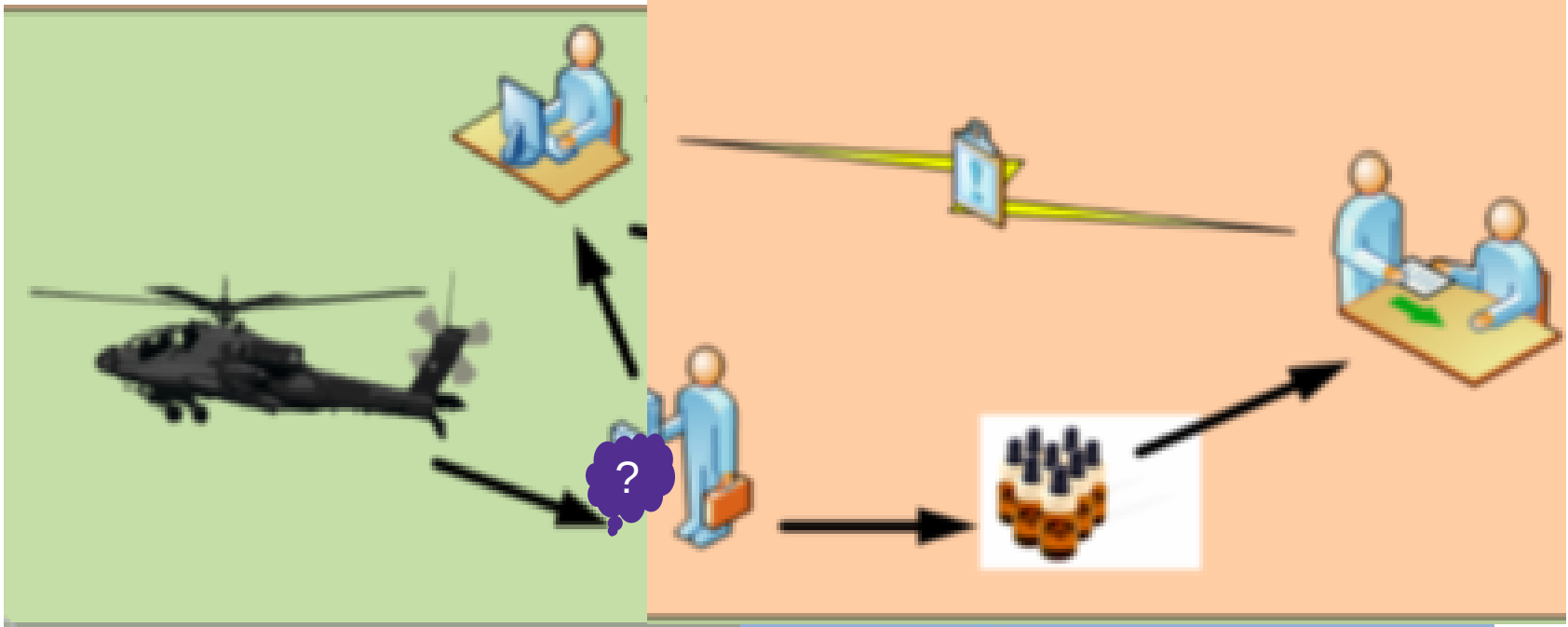
- Users need answers NOW
- Answers NOW are more valuable than later
- With lab-based analysis, users need to WAIT for test results and need to act before test results come back
- Sometimes it just too LATE
- User's of high value, mobile assets need real-time portable CBM



The importance of NOW

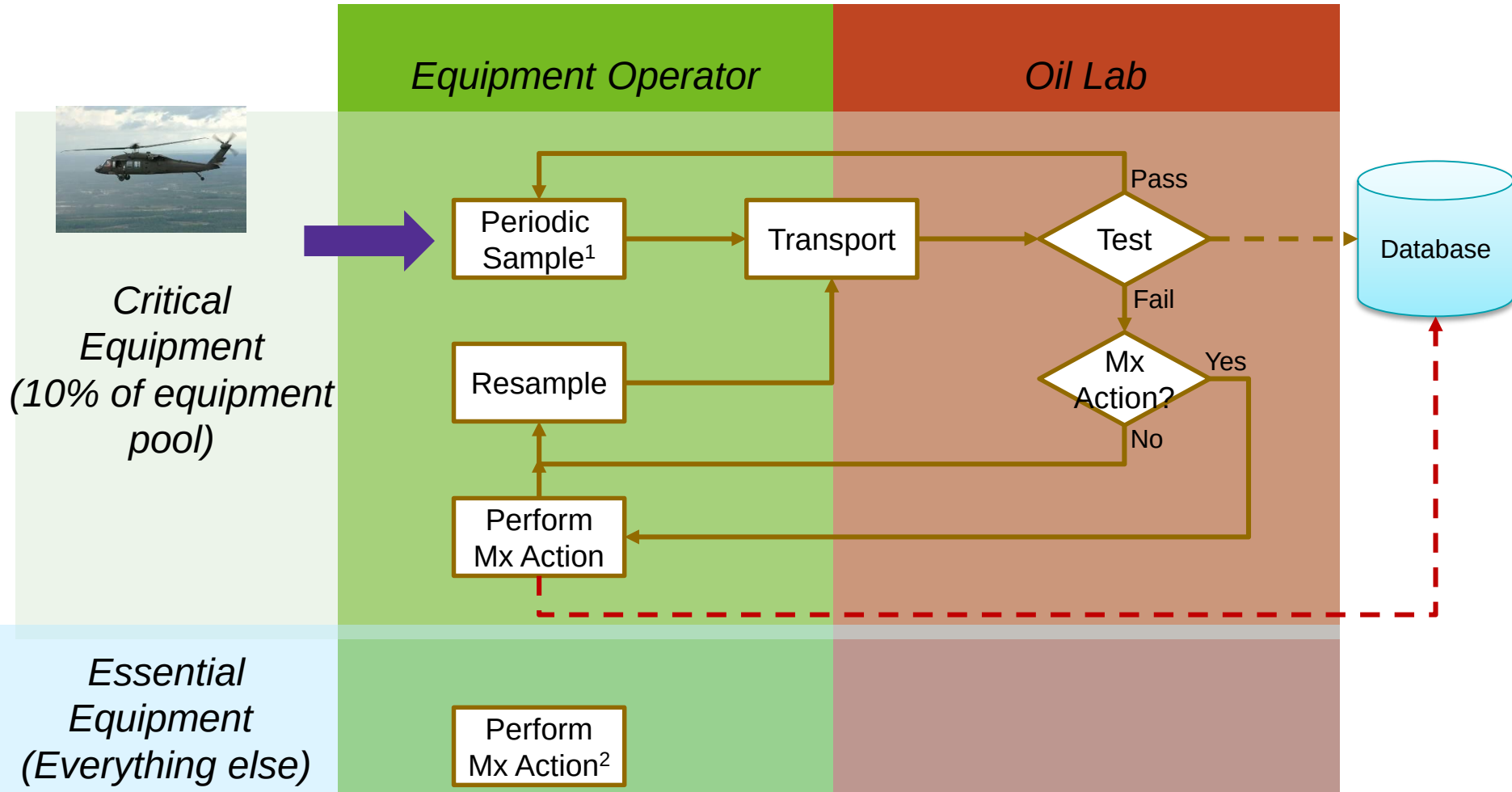


The Value of Getting Answers Now



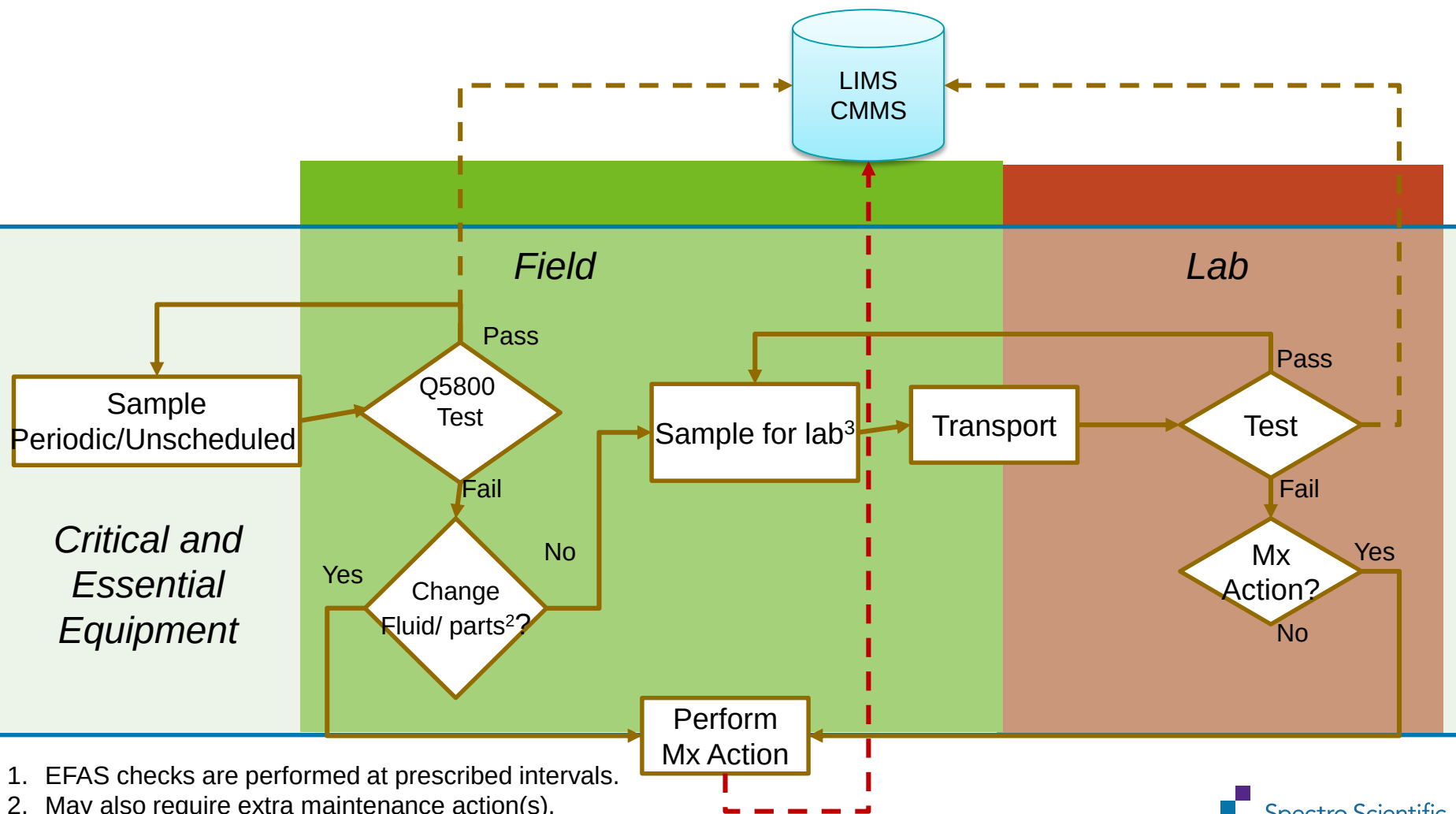
- Sample at the asset
- Go/No go on fluid quality and contamination
- Detect severe wear if present
- Recheck there and then
- Reduce logistics trail

Current Oil Analysis Programs



1. Routine samples are taken at prescribed intervals or samples sent based on certain events (e.g., metal chunks in sump).
2. Fluids and filters are changed at prescribed intervals . No routine oil analysis done.

Expeditionary Fluid Analysis



1. EFAS checks are performed at prescribed intervals.
2. May also require extra maintenance action(s).
3. Samples are taken at prescribed intervals or when EFAS device results dictate.

Extending Oil Analysis in the field

Phase I: Fluid Analyzer

- Physical properties
- Cross Contamination
- Degradation & Water
- Flip-Top sampling

2009

Fluidscan



Phase II- Lubricant Condition

- Add Kinematic Viscosity
- Combo Kit

2011



Combo Kit

Phase III Lubricant and Machine Condition

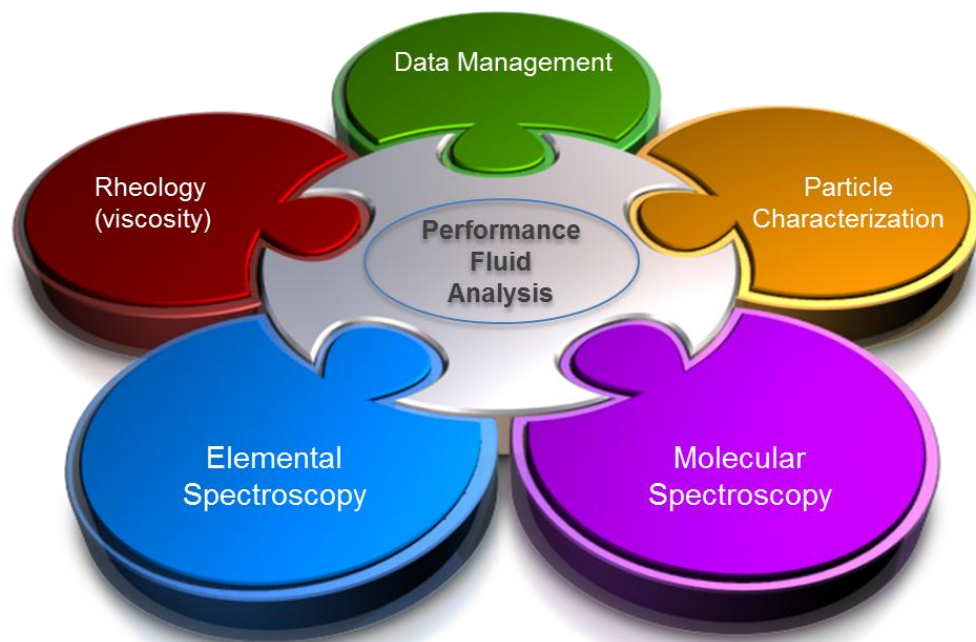
- Complete Integration
- Wear Debris Analysis Capability

2014



Q5800

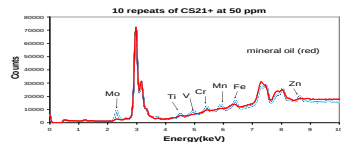
Q5800 Addresses all the key areas



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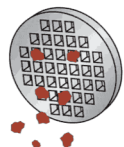
Results for Field Decisions



XRF Elemental Analysis

Silicon (Si), Aluminum (Al), Chromium (Cr),
Titanium (Ti), Iron (Fe), Nickel (Ni),
Lead (Pb), Copper (Cu), Tin (Sn),
Molybdenum (Mo), Silver (Ag),
Zinc (Zn), and Vanadium (V)

**ASTM WK38220 (in
committee)**



**Particle Count
Pore Blockage**
> 4 μ m

ISO 21018-3 (>4 μ m)

Patents granted : Fluidscan
Pending: Viscometer, FPQ approach



Infrared (IR) Spectroscopy

Physical Properties

TAN (Total Acid Number)
TBN (Total Base Number)
Oxidation, Nitration, Sulfation
Soot, Glycol, Water, Fuel
Incorrect Fluid
Antioxidant Depletion
Antiwear Depletion

ASTM D7889



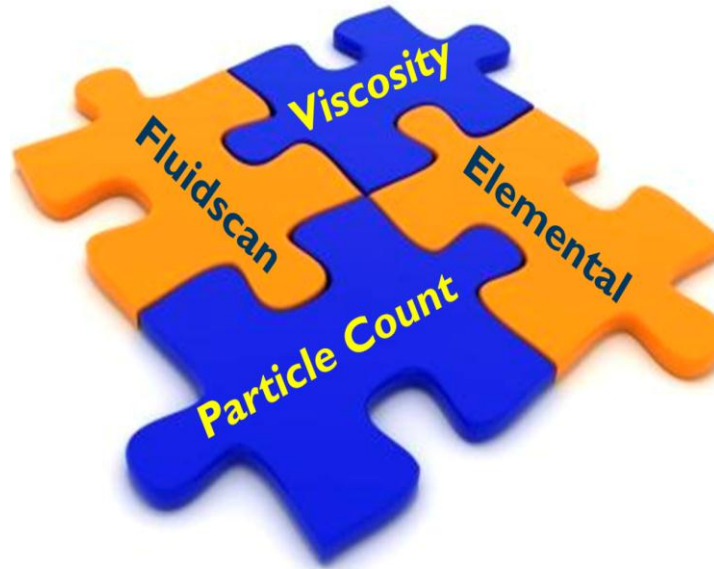
**Viscometer
Kinematic
Viscosity**
At 40° Celsius

ASTM WK40831 (in committee)



Why is it a Unique Solution?

- Portable
- Solvent Free
- All in one carry case



Patents granted : Fluidscan
Patents Pending: Viscometer, FPQ



PRODUCT INFORMATION

Part Number	Q5800 EFA
Applications	Mineral and synthetic lubricants including gear, engine, transmission, hydraulics, turbine as well as military, marine and mining applications.
Operating Mode	Single test or multi-parameter analysis
Output	
Elemental (ppm): (dependent on fluid type)	Silicon (Si); Aluminum (Al); Chromium (Cr); Titanium (Ti); Iron (Fe); Nickel (Ni); Lead (Pb); Copper (Cu); Tin (Sn); Molybdenum (Mo); Silver (Ag); Zinc (Zn); Vanadium (V)
Fluid Chemistry:	TAN & TBN (mg KOH/g); Oxidation, Nitration, Sulfation (Abs/.1mm); Water (parts per million); Glycol (% by weight); Soot (% by weight); Incorrect fluid (% by weight) Antioxidant Depletion (% remaining); Antiwear Depletion (% by weight)
Viscosity:	Kinematic viscosity at 40°C (cSt)
Particle Count:	Particles > 4 µ/ml on filtergram
Methodology	ASTM D7889; Mod ASTM D7279 (visc); Mod ISO 21018-3 (Particle Count)
Repeatability	Viscosity: +/- 5% Fluid Chemistry: per ASTM D7889 Particle Count: per ASTM D7467 Elemental: reliant on particle count
Calibration	Factory, Verification Standards: NIST traceable verification standards provided

Specifications

OPERATIONAL SPECIFICATIONS

Sample Volume Required (all tests)	Less than 5 ml per sample
Sample Time Required	Viscosity: 20 secs to 10 mins, dependant on grade Fluid Chemistry: less than 60 secs Particle Count: 20 secs to 3 minutes, sample dependent
Solvents/Reagents	None
Ambient Operating Temperature	-10° to +50°C (all modules w/o viscosity) With viscosity: 0° to 40°C
Operational Humidity	RH< 80% non-condensing



Portability

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USER INTERFACE SPECIFICATIONS

Software/Operating System	Windows® CE (built-in)
Display	Fixed angle color touchscreen display
Data Storage	Internal flash memory (SD Card Expansion)
Data Transfer	Ethernet, USB
Security	Password protected
Data Entry	Touchscreen/ Desktop Software (Asset loading)
Communication	WIFI (optional); Bluetooth (optional)

POWER REQUIREMENTS

Battery Power Source	Exchangeable lithium-ion battery pack
Charge Power	AC 110/240 V, 50/60 Hz, 10 watts
Typical Runtime	4-6 hours
Recharge Time	2.5 hours

MECHANICAL SPECIFICATIONS

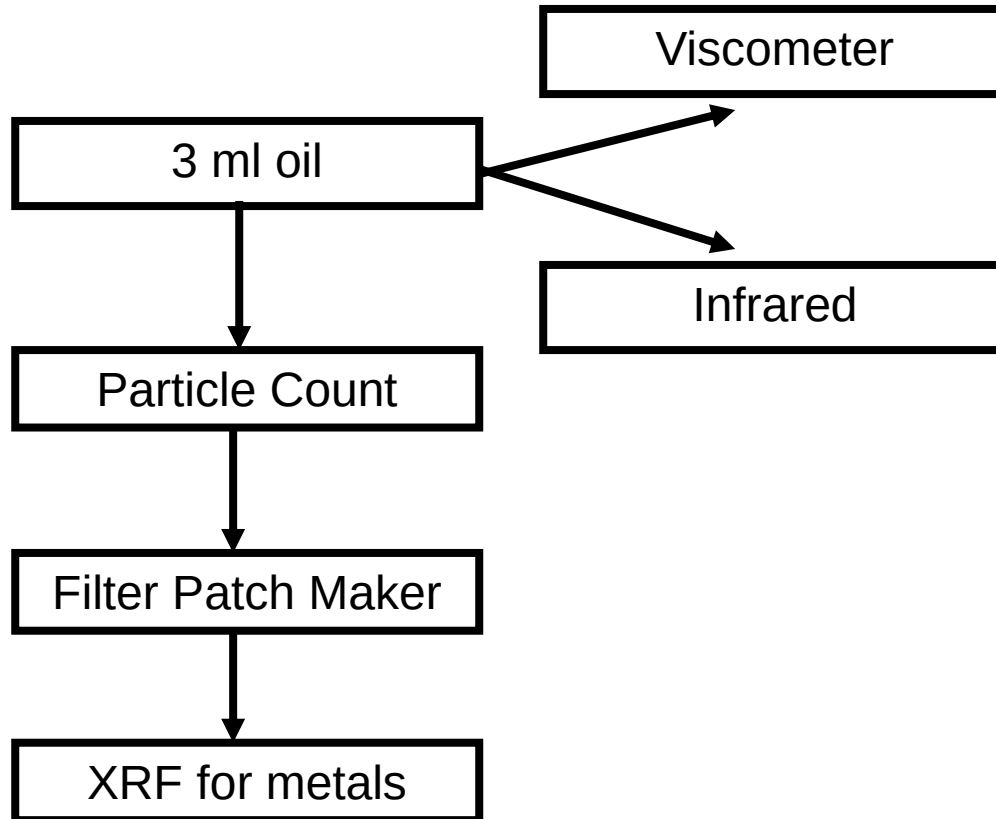
Dimensions	48 cm (L) x 39 cm (W) x 23 cm (H); 19.2" x 15.2" x 9"
Weight	16.5 kg (36.4 lbs); 20 kg (44.4 lbs with backpack and power supply)

COMPLIANCE

CE, RoHS, EN 60610-1:2010; 21 CFR part 1002



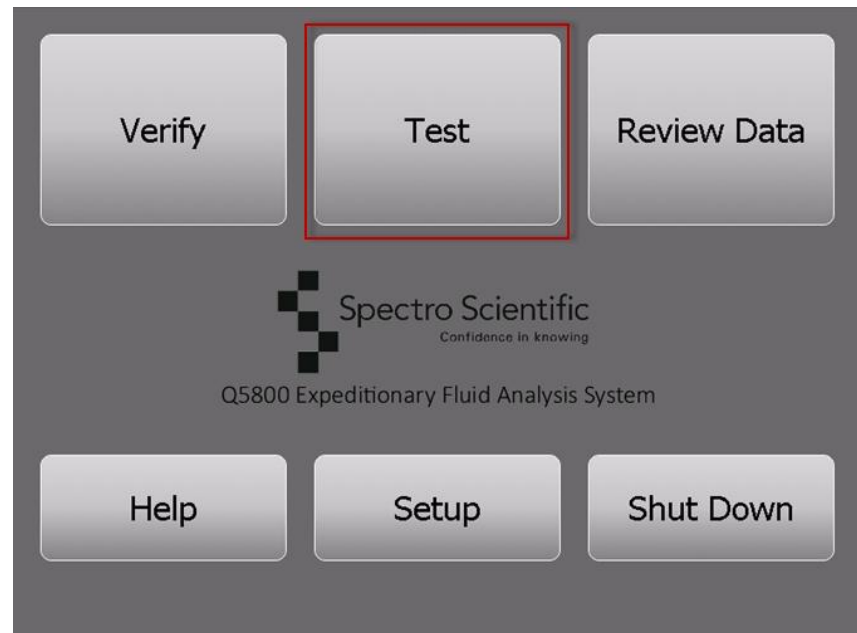
Q5800 – System Operation 3-7 Minutes



1. Draw 3 ml fluid
2. Place 2 drops in infrared
3. Put 2 drops in viscometer
4. Concentrate Sample on filter
5. Place coupon in XRF
6. View Analysis, Wipe Clean

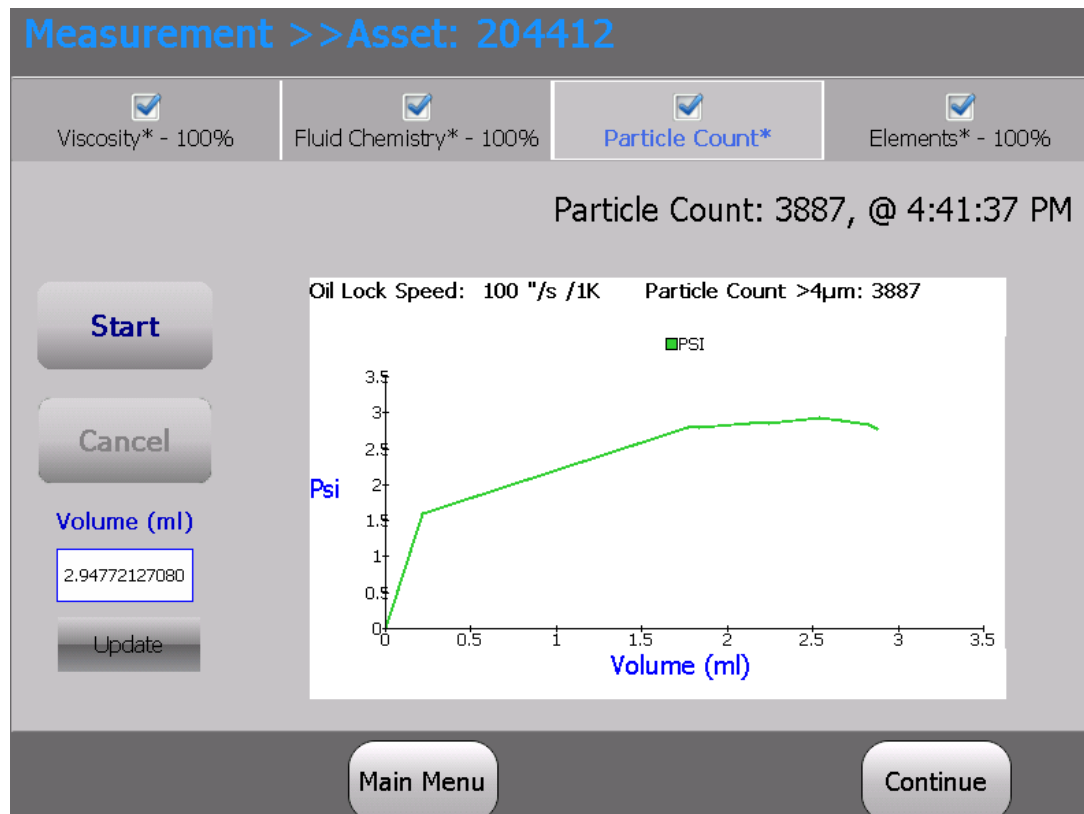
Instrument Controller Software

- Simplified Process
- Windows touch screen
- Asset-driven with trending histories
- Results are synchronized to Fluid Manager Desktop Software



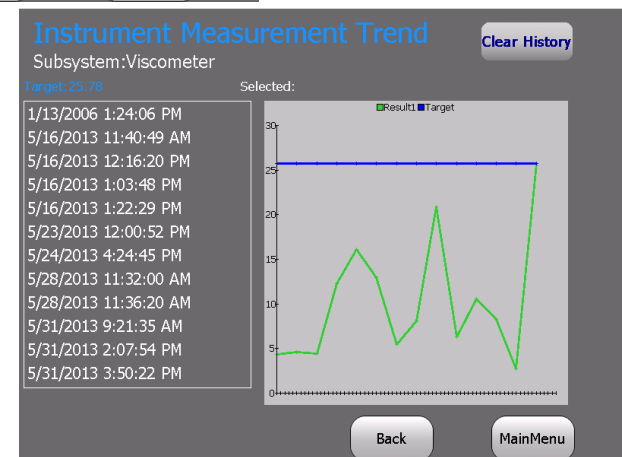
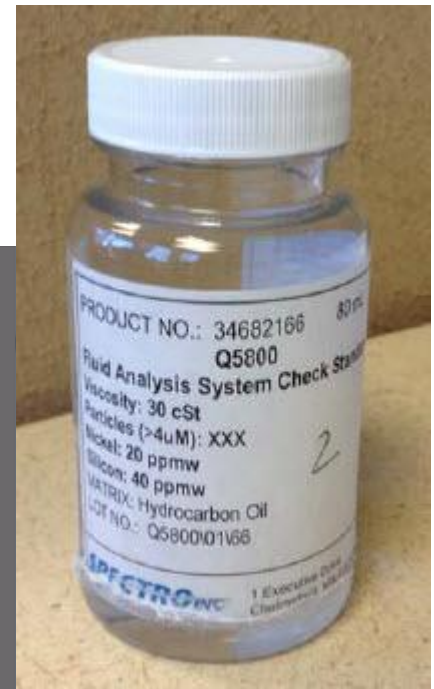
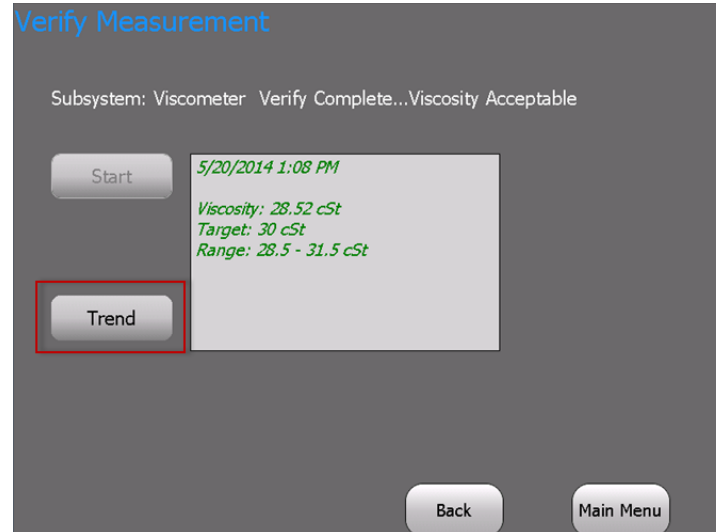
Q5800- Running a Sample

- Tabbed screens
- Simultaneous operation



Verify Calibration

- Multi parameter standard
- Advises operator of pass/ fail
- Provides basic QC trend



Results

- Results all together
- Alarms employed (if set)

ReportNewScreenAsset: 204412

10:40



IR Properties  

AW Additive	%	0	9.90
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Viscometer Properties

Visc-40	Cst	200	96.30
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FPQ Properties  

Amt Dispensed	ml	0	2.87
Particle > 4m	#/ml	200	3887.42
Particle > 4m	#/ml	200	3887.42

XRF Properties  

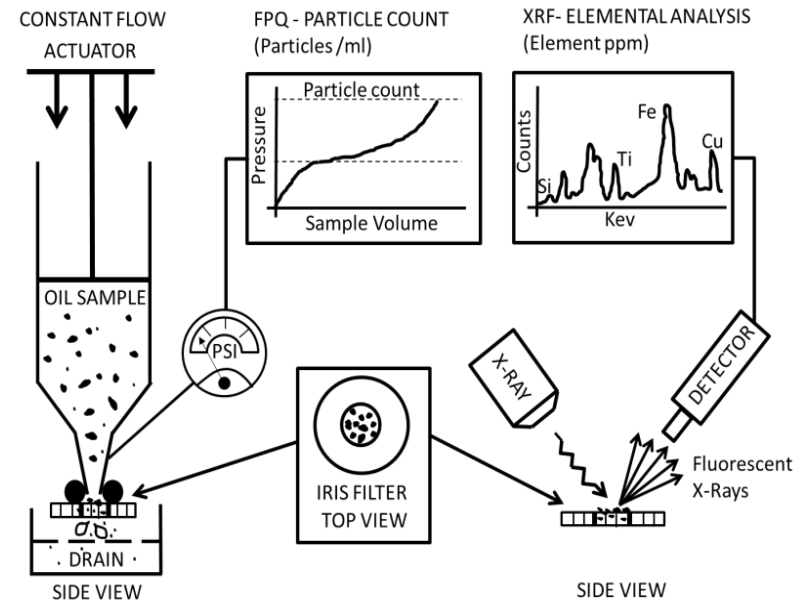
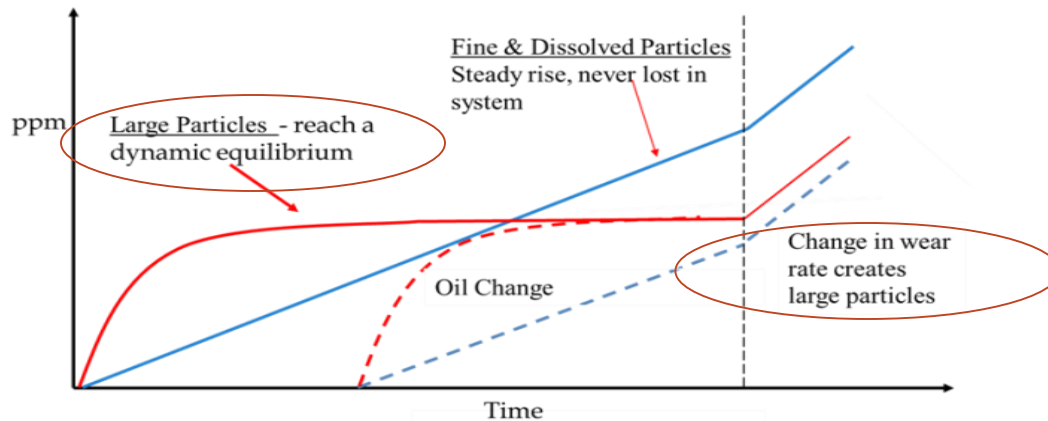
Ag	ppm	100	0.00
Al	ppm	100	0.00
Cr	ppm	100	0.00
Cu	ppm	100	0.00
Fe	ppm	100	0.00
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Save

Technical Discussion- Understanding the Particle Count/ Metals Output

Understanding the FPQ/XRF output

- Results will
 - AGREE with Ferrography for Abnormal Wear
 - CORRELATE to RDE Spectroscopy for wear conditions where both large and small wear particles are trending to failure
 - WILL NOT
 - Detect dissolved material /additive elements quantitatively



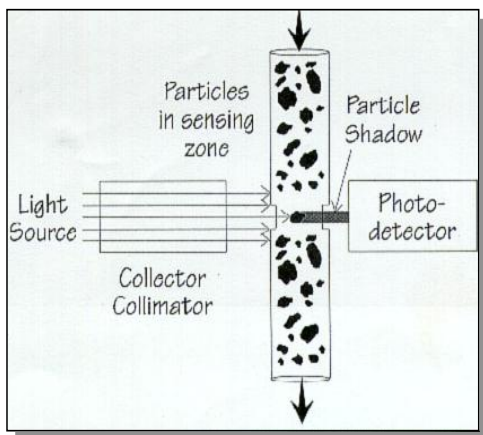
If there is abnormal wear debris present, Q5800 will catch it

Machine Condition Assessment

- Particle Count & Elemental Quantification
- How many and where is it coming from?
- Existing Techniques
 - Particle Count
 - Laser Counters
 - Direct Imaging
 - Pore Blockage
 - Elemental Quantification
 - EDXRF
 - Atomic Emission Spectroscopy (ICP, RDE)
 - Microscopy / Ferrography

Particle Count Techniques

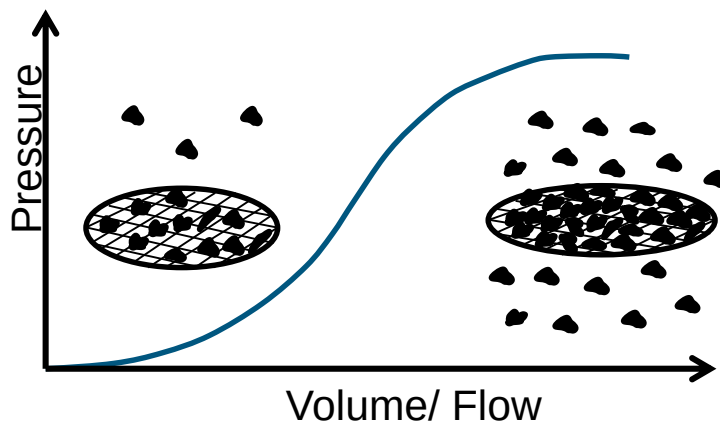
Light Blocking



Ideally suited to hydraulics and clean systems

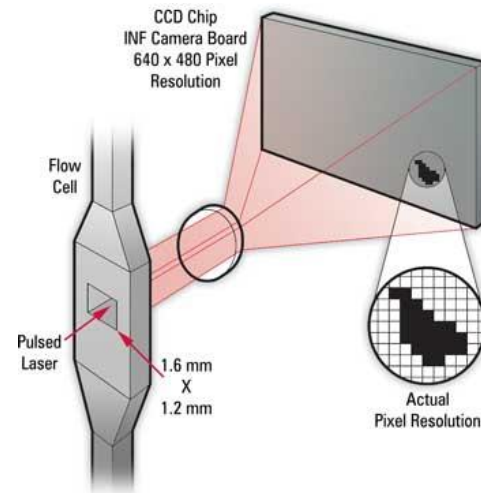
Pore Blockage

Mesh obscuration



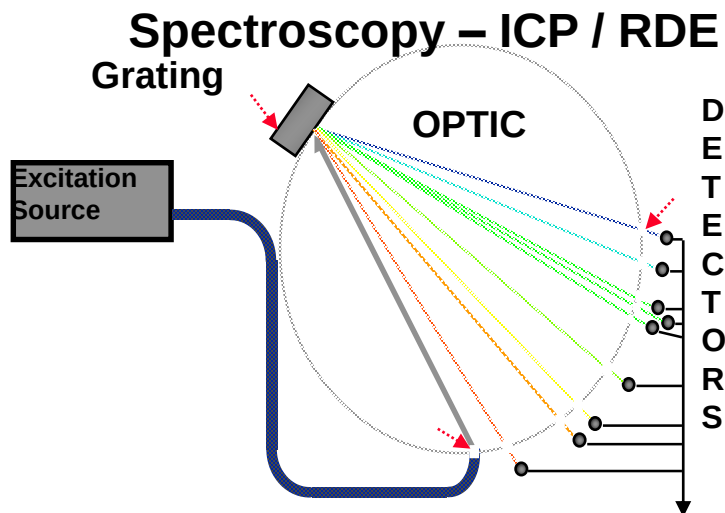
No interference from soot, water and other soft particles

Direct Imaging

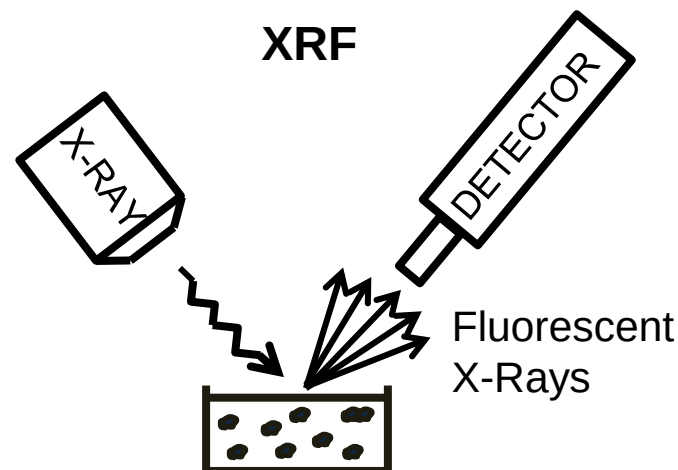


Ideally suited to systems that generate wear naturally i.e. diesel engines, can tolerate some soot

Quantitative Elemental Techniques



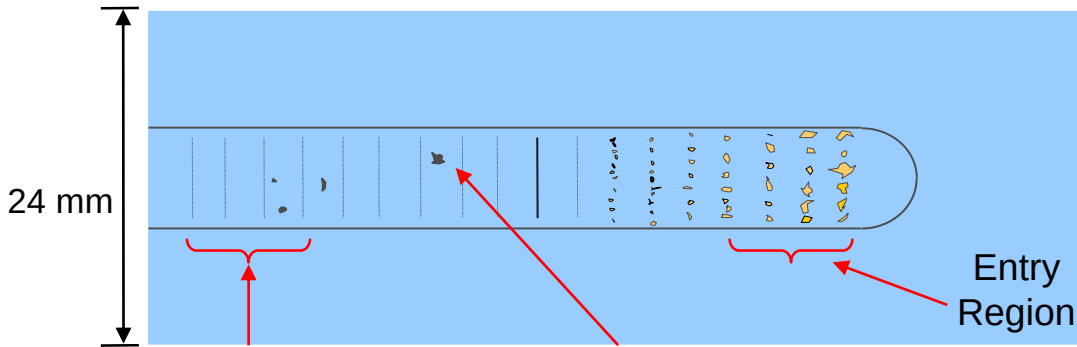
- Very poor large particle Sensitivity
 - 5um and above little detection
 - Data is trended using small particles only



- Good Large Particle Sensitivity depending on sample presentation
 - Cup produces poor sensitivity
 - Concentrated particles provide excellent sensitivity

Microscopy – Ferrography/ SEM EDX

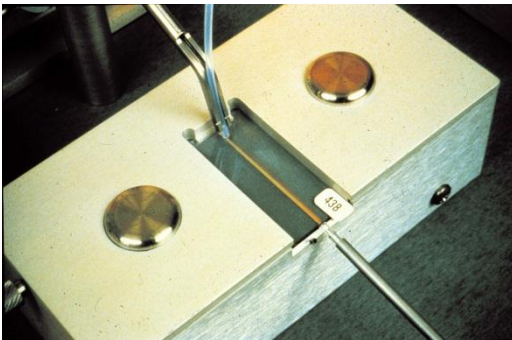
Ferrogram Substrate



Ferrous Particles Near Exit are Submicroscopic

Non Ferrous & Weakly Magnetic Debris Deposited Randomly

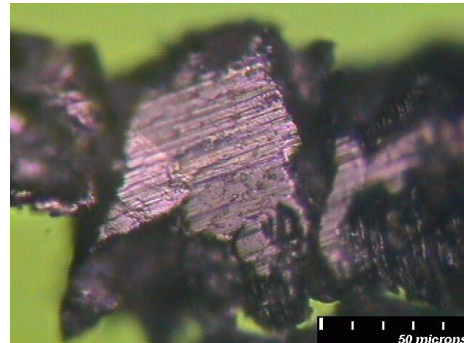
Ferrogram Maker



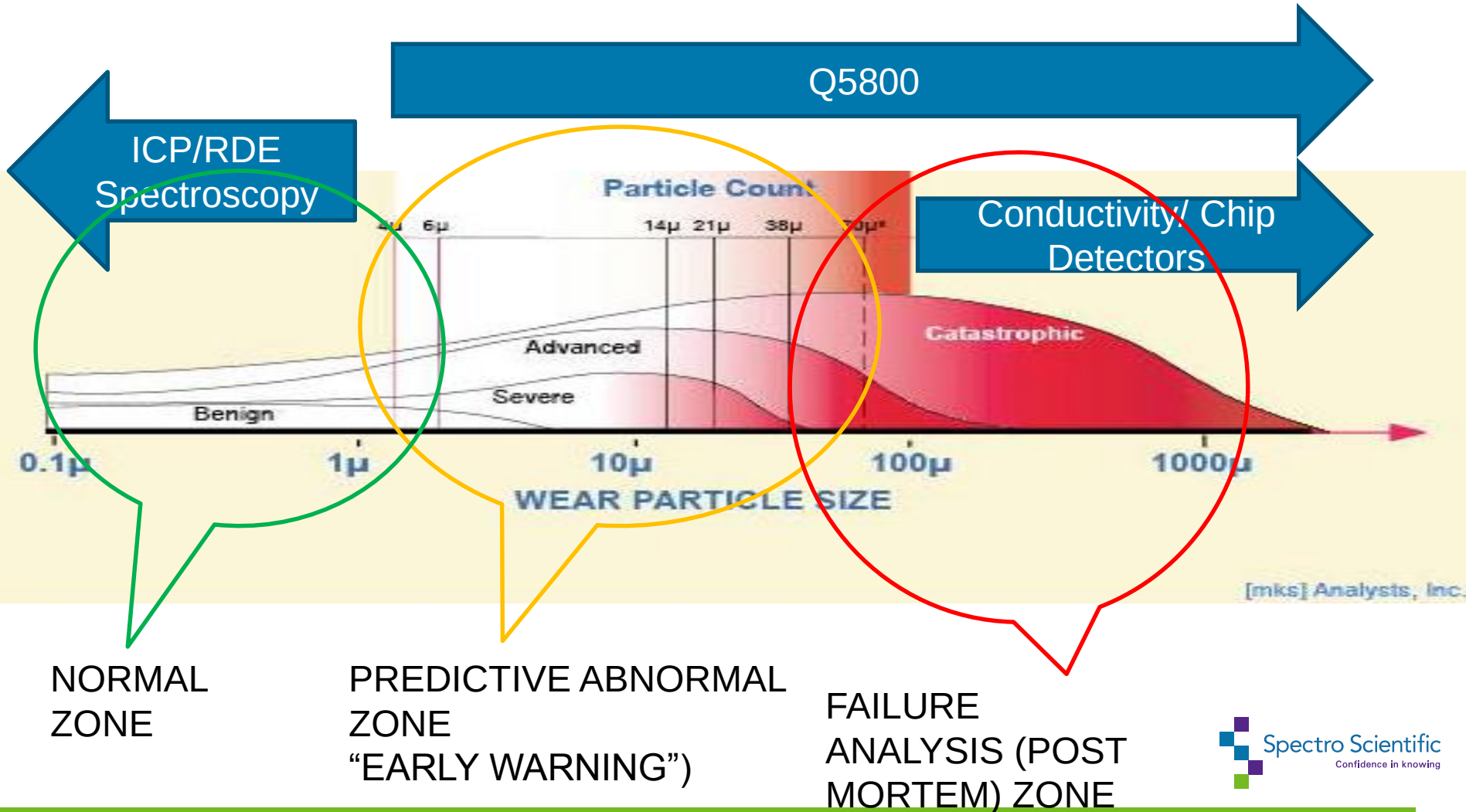
Optical Microscope



SEM EDX

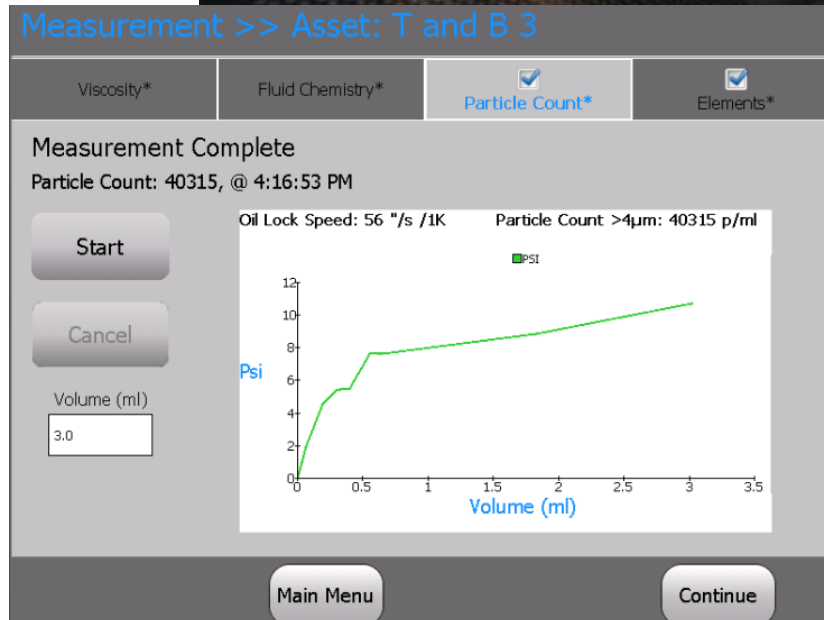


Wear Particle Size Detection Zones

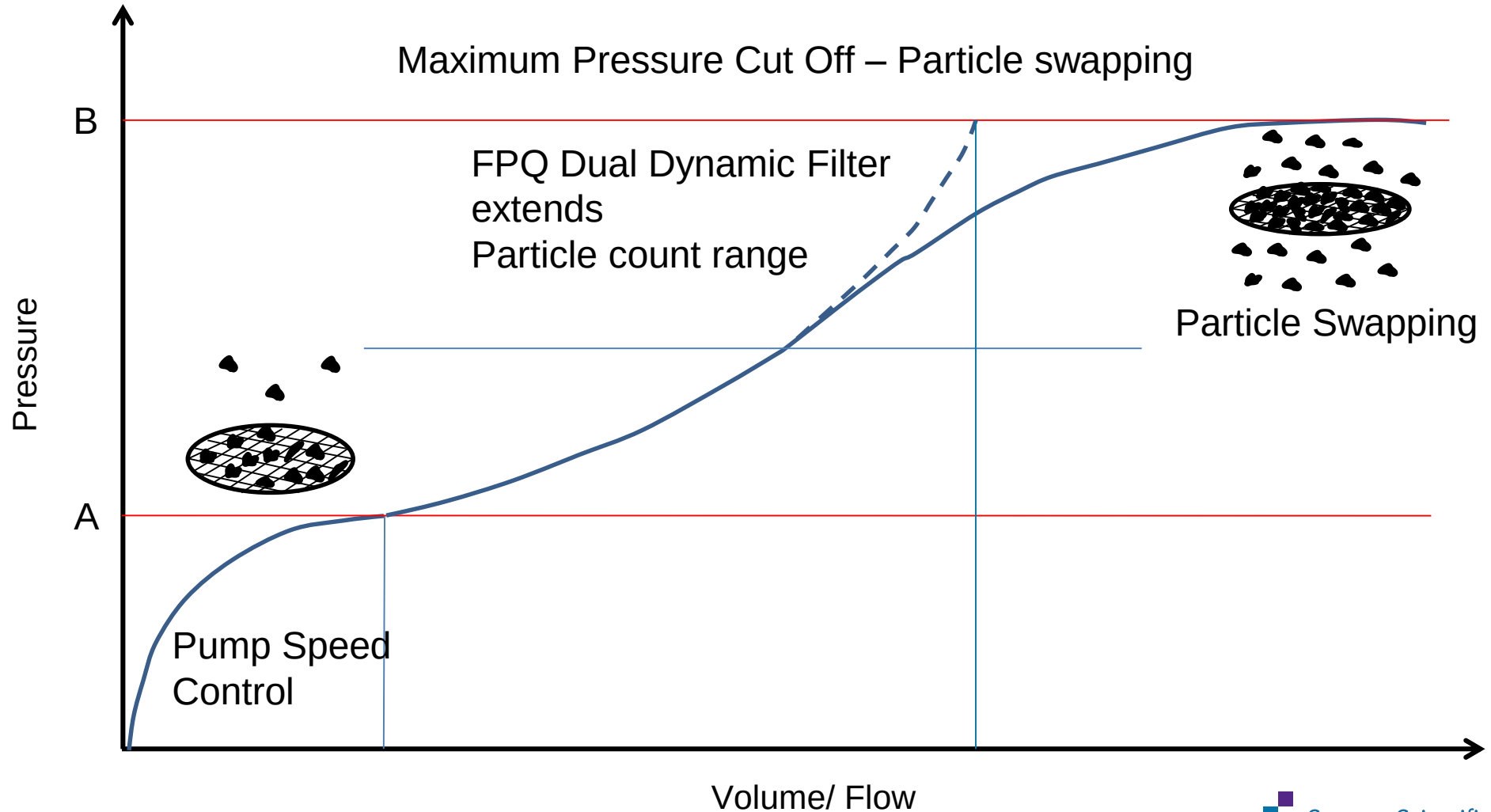


FPQ Module

- Single Channel Particle Count $>4\mu\text{m}$
- Calibrated from 5,000- $\rightarrow$ 1,000,000 p/ml
- ISO >4 Code 19 to 27 Range
- Viscosity: 5cSt- 350cSt- calibrated range
- Analysis time 30 secs $\rightarrow$ 5minutes depending on oil viscosity and particle quantity
- No soot or soft particle problems

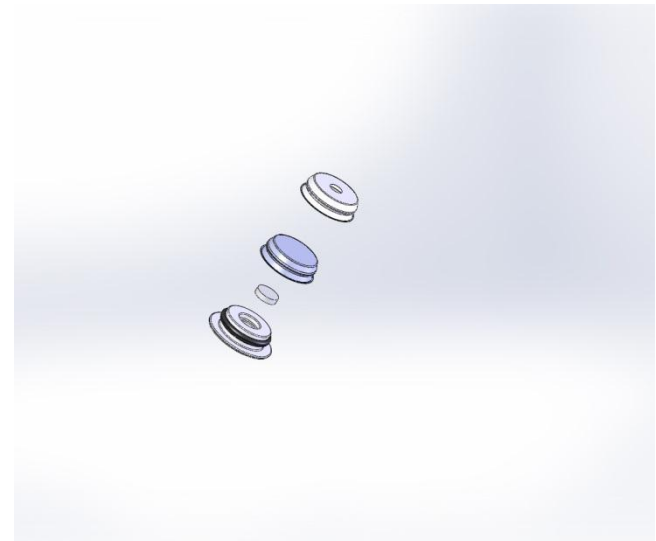


FPQ Filter vs. Conventional Pore Blockage Filter



FPQ Filtergram

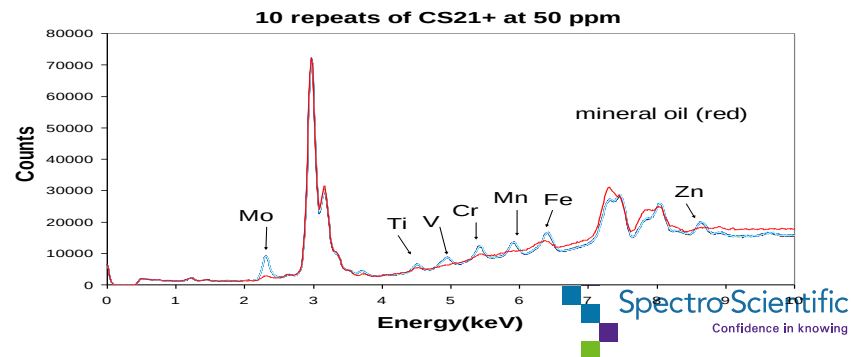
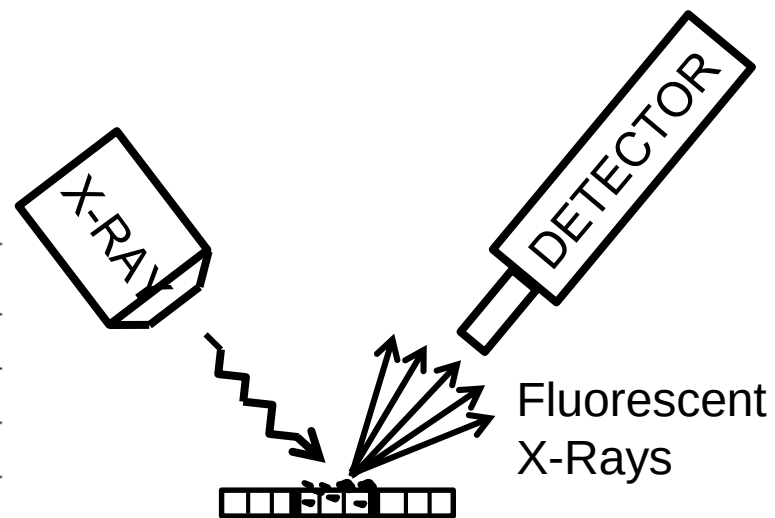
- Unique IRIS Corralling Feature for particle laden oils
- IRIS is 1/8" (3.175mm) diameter
- Uses a patented dual dynamic Polycarbonate film filter (PCTE), ~ 30,000 pores 5µm diameter holes
- Combination filters
- Avoids traditional caking



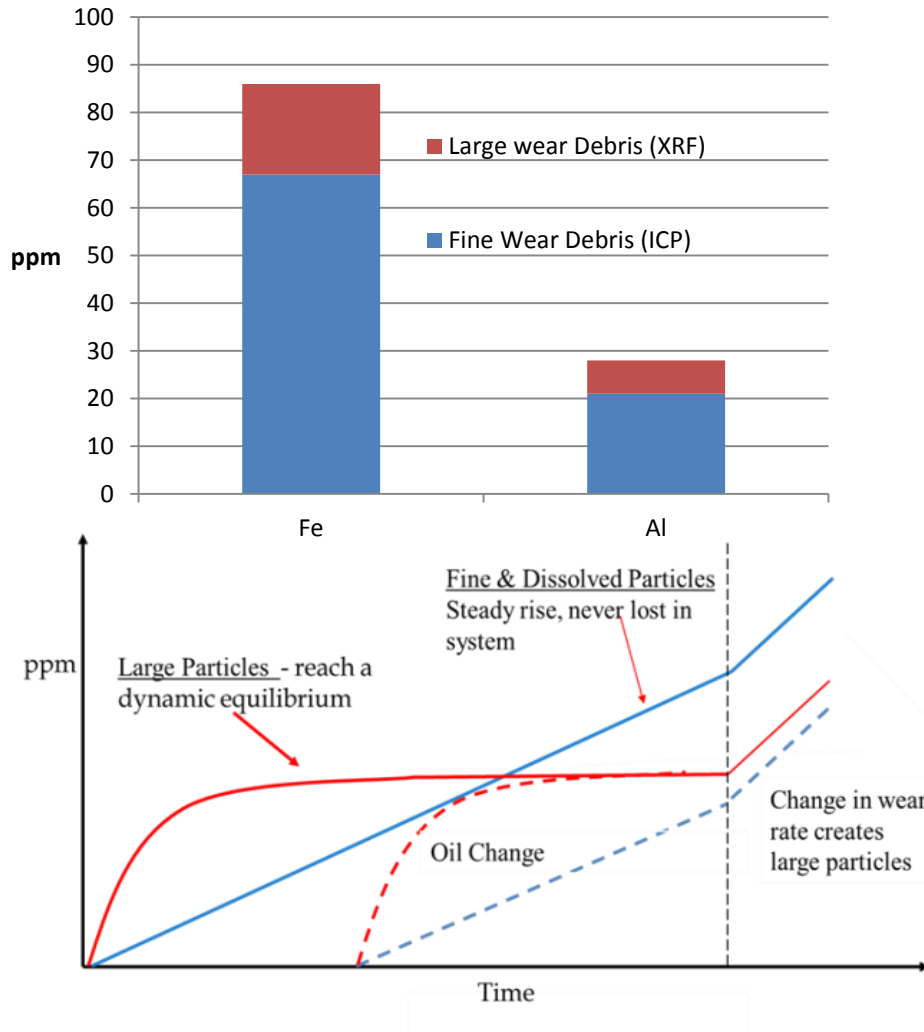
XRF Module

ELEMENTAL MODULE

Detector	10 mm ² SDD Detector; peltier cooled
Resolution	145 eV at 100,000 Cps
Excitation Source	X-ray tube with Rhodium target; max voltage 45kV



Large Particle Distribution



- Large Particle quantities are low compared to quantities of finer debris detected by ICP
- Few Boulders, lots of sand concept
- Consistent with findings using Acid Digestion
- Large particles reach dynamic equilibrium
- Small particles steadily rise

How influential is a large particle?

Wear Particle Weight

$$V = \pi d^2 h / 4$$

$$V = \pi (20 \times 10^{-4})^2 (4 \times 10^{-4}) / 4$$

$$V = 1.26 \times 10^{-9} \text{ cm}^3$$

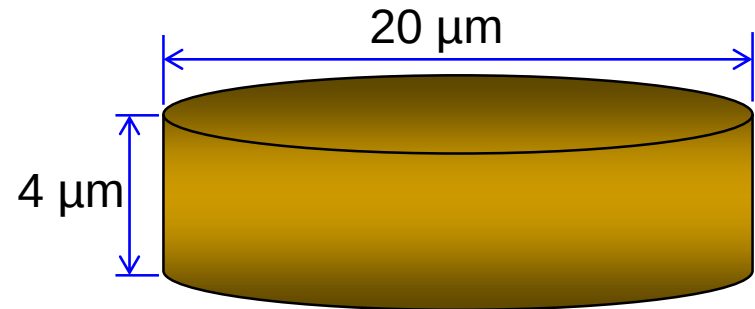
$$\text{Weight} = \text{Density} \times \text{Volume}$$

$$\text{Weight} = 8 \text{ g/cm}^3 \times 1.26 \times 10^{-9} \text{ cm}^3$$

$$\text{Weight} = 10 \times 10^{-9} \text{ g} = 10 \text{ ng}$$

PPM (mass) vs. Particle
concentration (quantity) on the FPQ
filter

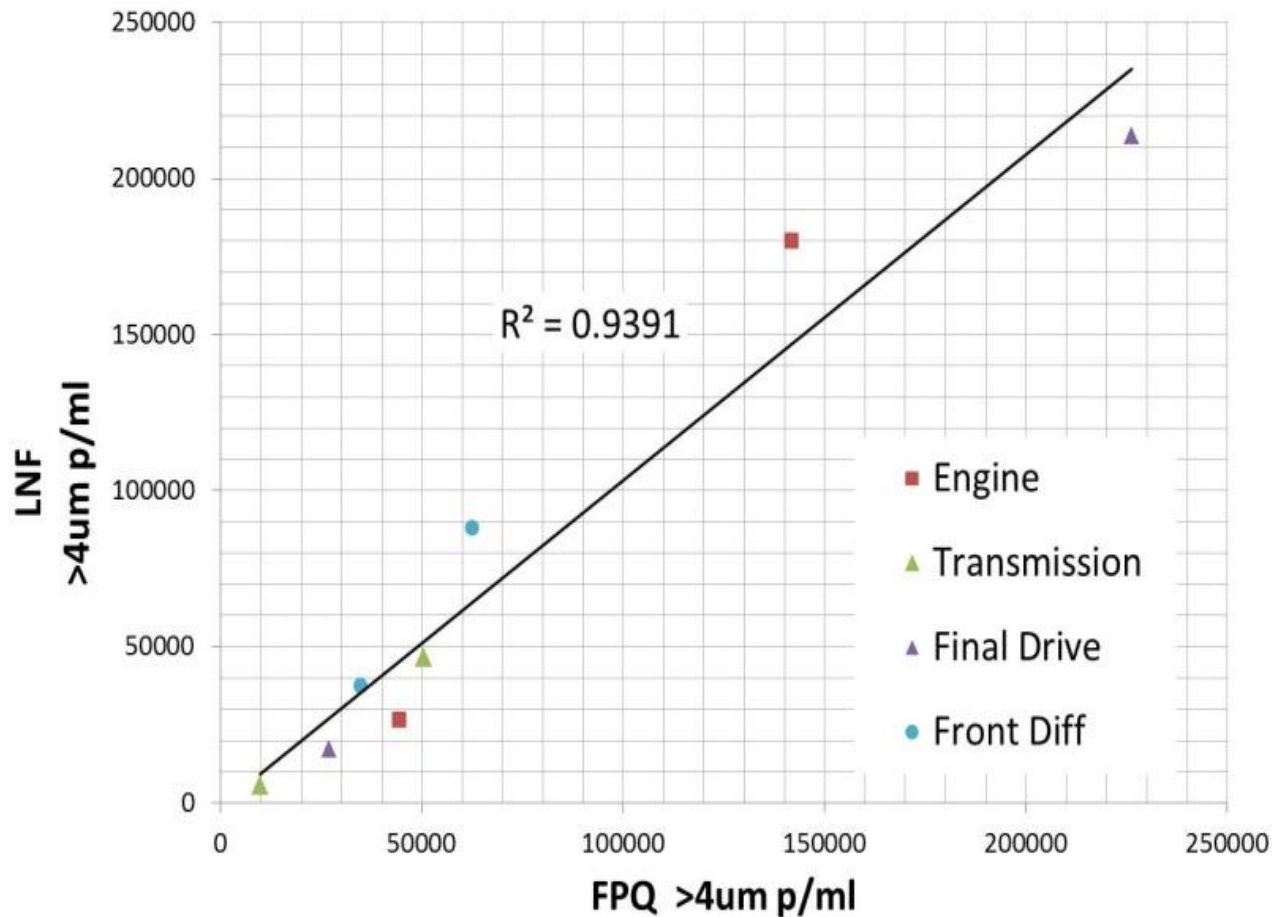
Just 30 of these could
indicate bearing failure!



“Based on weight, 1 ml of oil would
need 100 particles similar to the
platelet above to increase the
concentration by 1 ppm

$$\begin{aligned} 100 \text{ particles as above} &= 100 \times 10 \text{ ng} \\ &= 1,000 \text{ ng} \\ &= 1 \mu\text{g} \end{aligned}$$

LNF vs FPQ for Different Applications

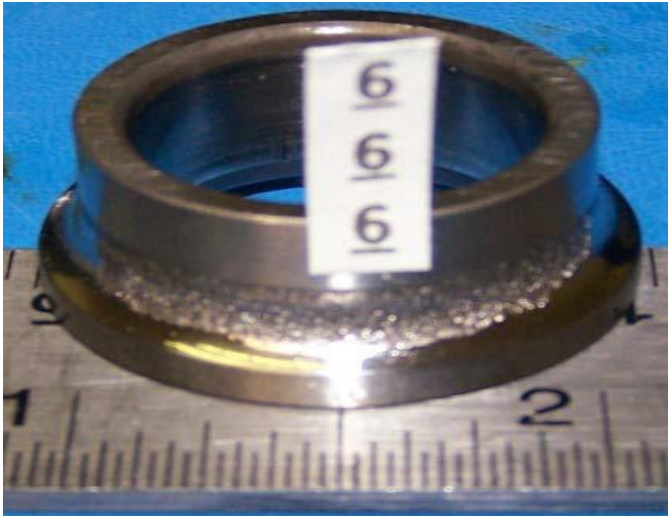


FPQ & XRF Data - Different Applications

Sample	Particles >4um (/ml)		Application	ITS Q 5800 XRF (ppm)			
	LaserNet Fines	FPQ data		Al	Cu	Fe	Si
E1 High wear	180209	141795	Engine	2.0	0.0	0.8	1.4
E2 Low Wear	26802	44188	Engine	0.5	0.6	0.6	0.7
T1 High Wear	46618	50390	Transmission	0.4	2.2	2.2	1.7
T2 Low Wear	5346	9664	Transmission	0.0	0.0	0.2	0.3
F1 High wear	213674	226222	Final Drive	4.3	0.0	8.4	7.0
F2 Low Wear	17185	26948	Final Drive	0.1	0.0	2.0	0.5
D1 High Wear	88193	62259	Front Diff	1.2	0.0	4.2	1.9
D2 Low Wear	37613	34773	Front Diff	0.9	0.7	2.9	1.2
E3 High Water	1025329	31686	Engine	0.5	0.0	1.0	0.4

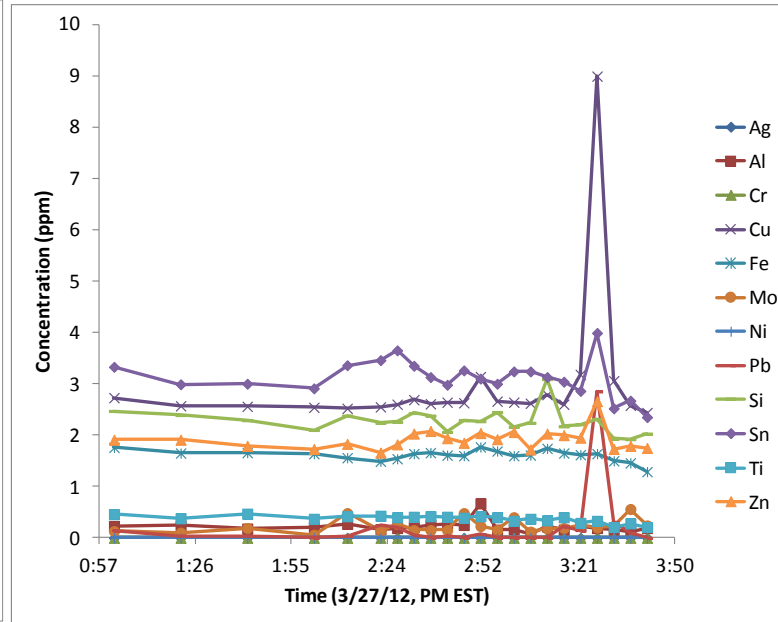
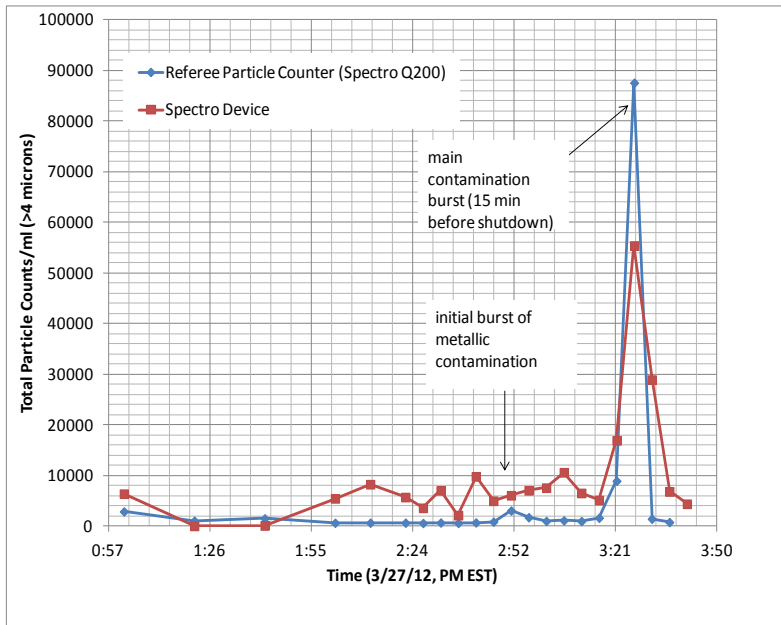
Applications Testing – Bearing Failure

Property	Aeroshell 500 Typical Specification	Sample 53	Sample 62	Sample 71
Viscosity (cSt, 40 C)	25.26	26.2	27	26.5
TAN (mgKOH/g)	0.01 (1 maximum)	0.25	0.23	0.2
Water (ppm)	n/a	982	1020	1054
Antioxidant Depletion (%remaining)	100	91	91	92



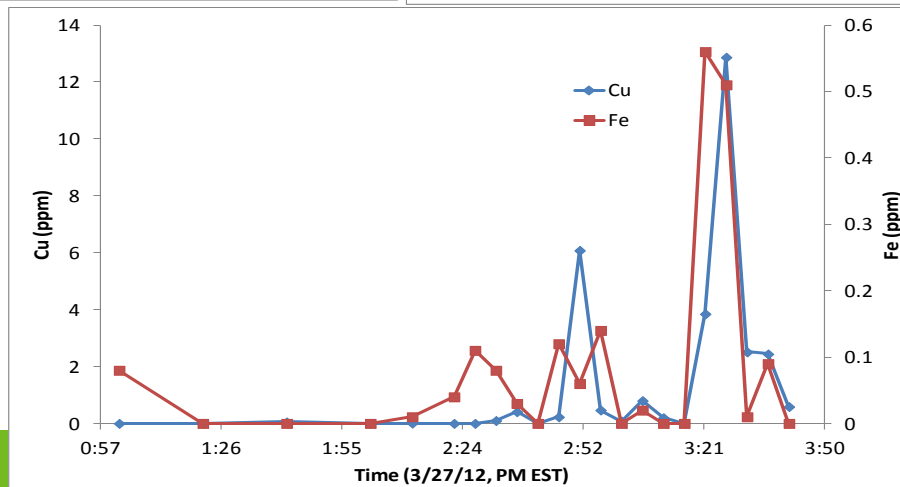
- Performed bearing accelerated lifetime testing
- Monitored with Q5800
- Results Demonstrate Capability of Detecting System Faults

Bearing Failure over a controlled period



RDE
Result

FPQ Particle
counter >4 μ m



Q5800
Elemental
Analysis

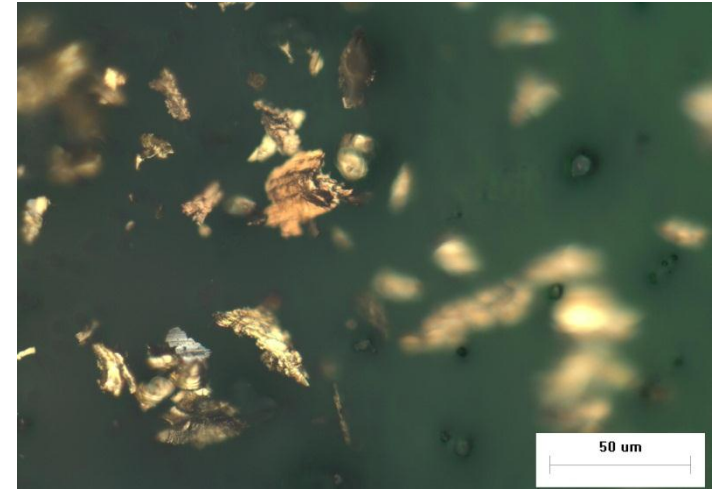
Case Examples

Q5800 detects abnormalities- Gearbox

Element	Result (ppm)
Al	0
Si	0
Ti	0
Cr	0
Fe	0.32
Ni	0.15
Cu	5.34
Zn	0
Pb	0
Mo	0
Ag	0
Sn	0



FPQ
Particle Count > 4
micron: 32172
(ISO 22)



Wear Debris on filtergram
patch reveals severe ,
recently generated copper
sliding wear
justifying an inspection of
gearbox

Heavy Truck at Auction Spot check

(E [REDACTED])

Specifications and Equipment Image

Equipment: 2002 Peterbilt 378 Truck Tractor
Engine: Caterpillar C12
Location: [REDACTED] Auction
Date: 5/30/2013
RB Lot: 289

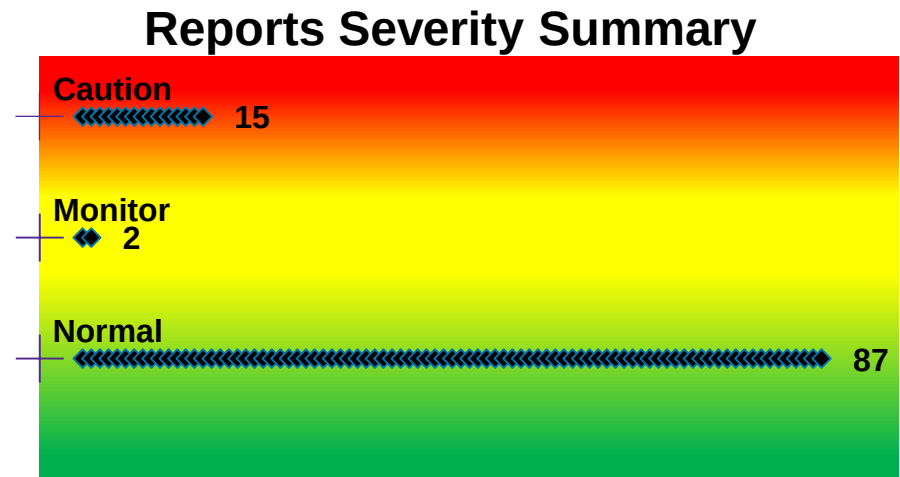


LAB #	SAMPLE DATE	WEAR / CORROSION INDICATORS										CONTAMINANTS						
		ALUMINUM: Pistons, cases, impellers, retainers	NICKEL: Gears, shafts, bearings	CHROME: Liners, bearings, rings, retainers, shafts, valves	COPPER: Bearings, bushings, washers, retainers, oil cooler	IRON: Gears, bearings, shafts, valves, liners, rings	LEAD: Bearings, bushings, oil cooler solder	MOLYBDENUM: Bearings, rings, oil additive	TIN: Bearings, bushings, oil cooler solder	TITANIUM: Gears, shafts, bearings	SILVER: Bearings, bushings, seals, oil cooler solder	SILICON: Breather, vent, turbo	FUEL DILUTION (%): Leaking injectors, blow-by	WATER (%): Condensation, coolant ingestion	ANTI-FREEZE (POS/NEG): Coolant Ingression	SOOT: Fuel system settings, excessive idling	PARTICLES (ISO Class): Total particulate level >4 Um	
289	5/28/2013	Δ	Δ	Δ	9	204	2	Δ	Δ	Δ	Δ	Δ	Δ	POS	Δ	27		

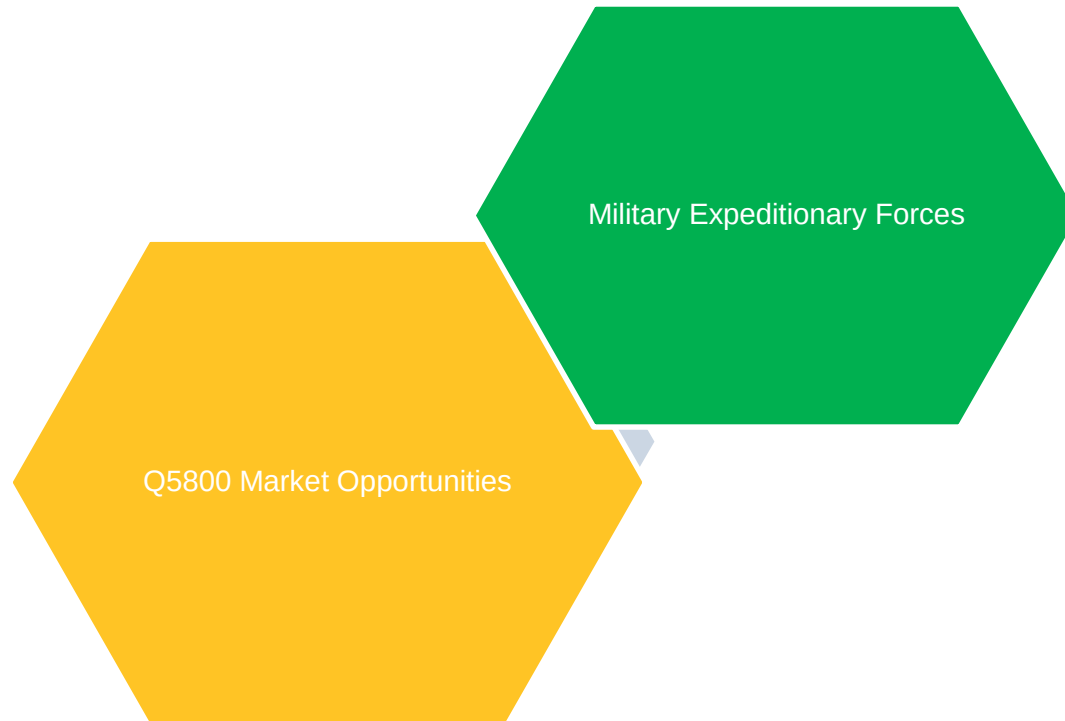
SEVERITY LEVEL CRITERIA

Heavy Equipment Auction Summary- Q5800 as a service tool

- Immediate answers= money
- Influenced buy/sell process
- \$75 per sample charged at site
(Customer said price too low per sample!)



Markets for Q5800





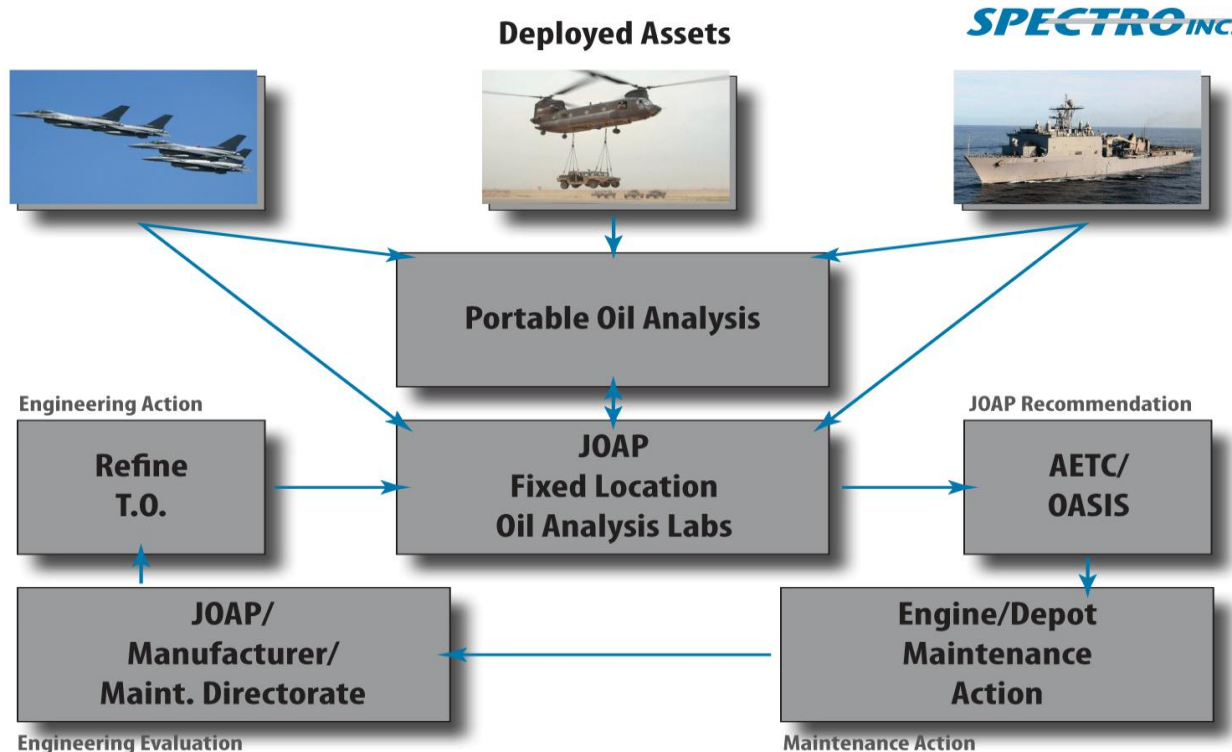
US Air Force Small Business Innovative Research (SBIR) Program

■ Phase I

- Proposal Submitted 01/2012
- Contract Awarded 02/2013

■ Phase II

- Proposal Submitted 02/2013
- Contract Awarded 04/2014





Project update: USAF

■ Current Challenges:

- Spectrometers can miss large particles
- Spectrometer transit case large and heavy
- SEM/EDX may also need to be deployed
- SEM/EDX approximately 50% effective
- JOAP guidelines need to be updated

■ Proposed Solution:

- Concentrate all particulate debris $>4\mu\text{m}$
- Eliminate matrix effects
- Produce true part per million concentrations
- Reduce logistics footprint for deployment
- Eliminate SEM/EDX requirement
- Create next generation CBM guidelines
- Establish closed loop maintenance feedback
- Refine guidelines and recommendations





Navy Expeditionary Combat Command

■ Current Situation

- Maintenance Management System requires oil change interval every six months
 - >30% of equipment remains in storage
 - Over 50% have low usage
 - Remaining equipment is run to failure
- New Regulations Mandate Annual 15% Fuel and Oil Reduction through 2018
 - Compliance Requires Implementation of Condition Based Maintenance Program

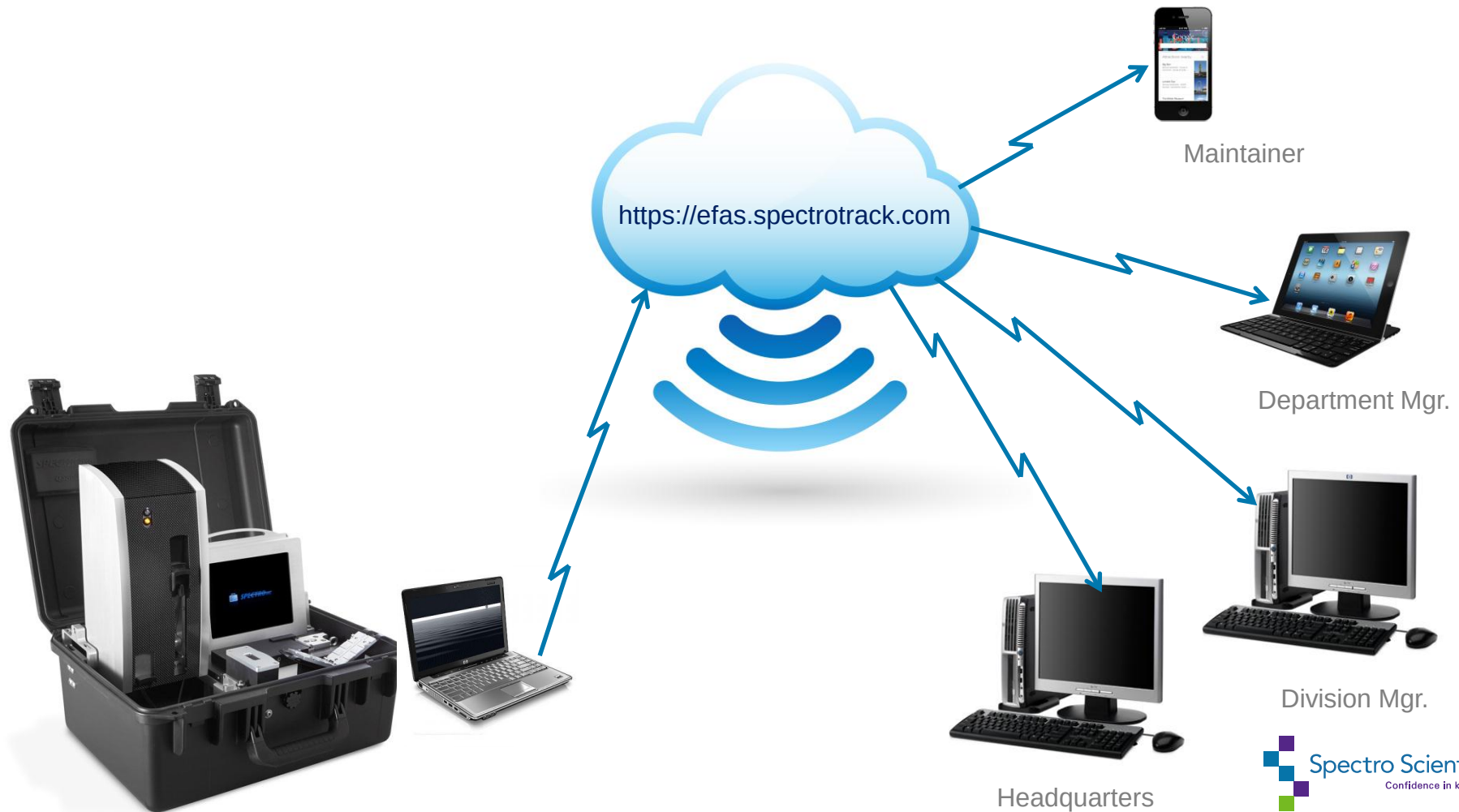


■ Solution Underway

- Develop Condition Based Maintenance (CBM) Program
 - Test and Evaluation Site Established in Gulfport, MS
 - Two Q5800's Installed and Maintainers Trained
 - Over 500 Assets Enrolled into SpectroTrack Database
 - Preliminary Maintenance Guidelines Established
 - Oil drain interval of >45% of assets can be extended.
 - Estimated annual cost savings/installation \$147,000

What's new- Q5800 Integrated with SpectroTrack

Complete Analytical Results at Point of Use



SpectroTrack NECC Fleet Asset Management

The screenshot displays the SpectroTrack NECC Fleet Asset Management web application. The interface includes a top navigation bar with links for Management, Sample Logging, Laboratory, Diagnosis, Administration, and Documents. A sidebar on the left shows a tree view of assets under the NECC category, with a list of various equipment including generators, trailers, and floodlights. The main content area displays details for a specific asset: "TRUCK TRACTOR MTRV MK31A1 W/O WINCH RTAA-Copy". This section includes fields for Name, Serial Number, Keyword, Manufacturer (MK31A1), Model (OSHKOSH TRUCK CORP), Asset Operating Time/Distance, and Location (NECC - P47). A large image of the truck is shown. Below this, two component details are listed: "Component: 9649843E" (ENGINE-C12) and "Component: 9649843T" (TRANSMISSION-4700 SP). Each component section includes fields for Component Type, Serial Number, Oil, Description, Last Sample, Samples Number, Serial Number (Sampling Port), Main Lab Report, and Limit. The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 10:11 AM on 4/6/2014.

TRUCK TRACTOR MTRV MK31A1 W/O WINCH RTAA-Copy

Name: TRUCK TRACTOR MTRV MK31A1 W/O WINCH RTAA-Copy

Serial Number:

Keyword:

Manufacturer: MK31A1

Model: OSHKOSH TRUCK CORP

Asset Operating Time/Distance:

Location: NECC - P47

Component: 9649843E

Component Type: ENGINE-C12

Serial Number (Component):

Oil: Raloy Raloy Diesel Supreme VDS-3 15W-40 CI-4/SL

Description: C12

Last Sample: **Samples Number:** 0 **Serial Number (Sampling Port):**

Main Lab Report: net.mecol.xman2.certificates.mecolCertificates.DefaultMecolCertificate **Limit:**

Component: 9649843T

Component Type: TRANSMISSION-4700 SP

Serial Number (Component):

Oil: Raloy Raloy Diesel Supreme VDS-3 15W-40 CI-4/SL

Description: 4700 SP

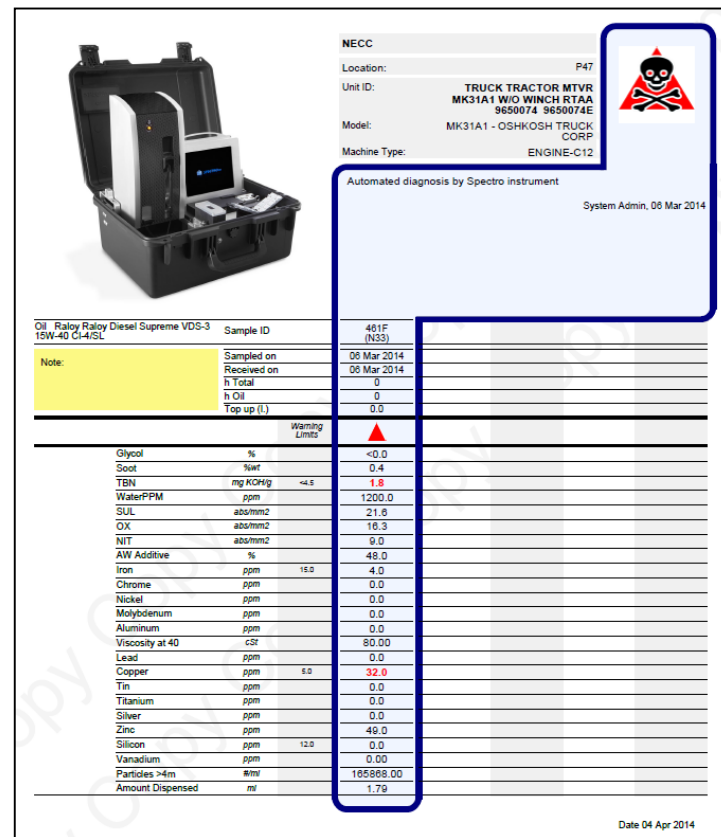
Last Sample: **Samples Number:** 0 **Serial Number (Sampling Port):**

Main Lab Report: net.mecol.xman2.certificates.mecolCertificates.DefaultMecolCertificate **Limit:**

Asset Running Frequency: periodically **Asset Running Description:** **Description:**

Fleet Condition and Maintenance Recommendations

- SpectroTrack Asset Condition Reports



Other Markets



Shell Offshore- Gulf Coast

Q5800

- Enables field oil analysis on rig
- Multiple assets can be tested on the one visit
- Data upload to Spectro Track for immediate review
- Units can be transported from rig to rig
- Increase value of technician visit



Expeditionary Oil Analysis Customer Persona: Mining Equipment

- Mine face equipment support
- Immediate result required while service truck is at equipment side
- Maintenance window limited
- Sample can be retained for central mine lab later

Tailor monitoring

- Investigate compartments for source
- Plan for parts exchange
- Confirm suspicions



How to Order

Product Information

- Website , Dropbox
- Data sheet
- White papers
- App note

- Demos units
 - Pooled
 - Plan demo with your regional sales manager

Products

Q5800 Portable Oil Lab

[Print](#)



The Spectro Q5800 is a rugged, portable expeditionary fluid analysis system that gives operators in the field the ability to perform comprehensive, mobile lubricant sampling. The battery-powered device enables complete lubricant assessment for condition monitoring and rapid results that permit informed maintenance decisions.

[Request Information](#)

- Features and Benefits
- Highlights
- Dimensions
- Related Products

Featured Documents

- Q5800 Expeditionary Fluid Analysis System
- A Comprehensive Solution for Expeditionary "On the Move" Oil Analysis
- Connecting Elemental Analysis to Particulate Count: A New Technique to Detect Failures

[View all documents](#)

Related Products



Spectro 5200 Trivector Analyzer



TurnkeyLab (TL)



FluidScan® G1000/PVisc Combo

What's new- Module Flexibility

- Modular Flexibility
- Q5800 EFAS
 - Complete Package
- Q5800 FPX
 - Wear metals and particulate only option
- Q5800-VFS
 - Lubricant condition only option



Integrated Solutions

	Portable Solution	Benefits
Root Cause Analysis	Mission Critical • Military • Mining • Marine 	Detects: • Element wear • Fluid chemistry • Fluid contamination • Comprehensive and integrated
On Site Screening	Fleet • Railway • Automobile • Offshore drilling • Marine • OEM Field Service 	Detects: • Fluid chemistry • Fluid contamination • Provides information on fluid condition to determine if an extended oil change interval is justified

Q5800

■ Improves maintenance execution

- Enables conditioned based maintenance (CBM)
- Enables real-time fluid assessment results, direction, and decisions
- Expands fluid assessment capabilities
- Reduces fluid changes based on hard-time intervals

■ Enhances the oil analysis program

- Without increasing the logistics footprint
- Reduces materiel and transportation costs (sampling equipment, bottles, shipping, etc.)
- Reduces processing samples that produce “normal” results
- Focuses laboratory operations on “abnormal” analysis and trending
- Supports an information-rich environment for improved decision making
- Allows laboratories to be more connected to maintainers

■ Technology Leadership





Thank You